

Satya Pradhan

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Smart Contract Security Researcher specializing in DeFi Protocols and EVM Architecture
Deep expertise in Code Review, Protocol-Level Threat Modeling, and Foundry.

Technical Skills

Security & Auditing: Manual Code Review, Threat Modeling, Invariant Testing, Stateful Fuzzing, PoC Development
Tools & Frameworks: Foundry (Forge, Anvil, Cast), Static Analysis (Slither, Aderyn), Hardhat
Smart Contracts: Solidity, Yul (EVM Assembly), Solana, Soroban, OpenZeppelin, ERC Standards (20/721/4626/4337)
DeFi Architecture: AMMs, Lending/Borrowing, Yield Stripping, Bridges, Cross-Chain Messaging (CCIP)
Languages: Solidity, Rust, C++, Python

Security Research & Audits

Independent Security Researcher

HackenProof, Sherlock, Code4rena, Cantina, CodeHawks, Immunefi

Bug Bounty & Audit Contests

02-2026 – Present

- HackenProof (Dual Defense): Awarded a paid bounty for discovering a Critical vulnerability in [Overlayer](#). Developed a detailed Proof-of-Concept demonstrating 'Cross-chain transfer lock via 'totalBridgedOut' underflow on destination credit'.
- CodeHawks: Secured Rank 28 out of 69 researchers in the [NFT Dealers](#) First Flight audit contest.
- Immunefi, Cantina, HackenProof: Submitted multiple valid duplicate findings across major bug bounty platforms, demonstrating a consistent ability to identify complex vulnerabilities in live production code.
- Routinely perform extensive manual and AI-aided code reviews and develop precise Foundry PoCs to exploit edge cases, state inconsistencies, and architectural logic flaws.
- Profiles: [HackenProof](#) | [CodeHawks](#) | [cantina](#) [Code4rena](#) | [Sherlock](#)

Smart Contract Security Training

Advanced Vulnerability Discovery

Cyfrin Updraft & CTFs

2025 – Present

- Completed advanced Smart Contract Security curriculum at Cyfrin, focused heavily on invariant reasoning, stateful fuzzing, and protocol-level threat modeling.
- Actively solve complex EVM attack vectors in Damn-Vulnerable-DeFi and Ethernaut CTFs to deepen understanding of low-level exploits.

Projects

Simple Account Abstraction (ERC-4337)

Solidity, Foundry, ERC-4337

[GitHub](#)

2025

- Implemented an ERC-4337 compatible smart account enabling gas abstraction and programmable validation.
- Engineered full UserOperation lifecycle validation, custom signature verification, and strict replay protection mechanisms.
- Simulated complex bundler flows and edge-case execution environments using Foundry integration tests.

AxoFi - Volatility Dam Protocol

Solidity, Mantle, EIP-4626

[GitHub](#)

2026

- Engineered a yield-stripping protocol separating Principal (PT) and Yield (YT) tokens, integrating Ondo USDY.
- Designed precise yield accounting and reward distribution logic, ensuring secure composability via the EIP-4626 vault standard.

Cross-Chain Rebase Token

Solidity, Foundry, Chainlink CCIP

[GitHub](#)

2025

- Built a multi-chain token utilizing Chainlink CCIP for programmable cross-chain transfers.
- Implemented secure burn-and-mint supply synchronization across chains and performed extensive fork-based simulation testing.

Education

IIIT Naya Raipur

B.Tech in Computer Science and Engineering

Naya Raipur, India

Expected 2027

- Relevant Coursework: Cryptography, Distributed Systems, DSA, Software Engineering, Operating Systems

Cyfrin Updraft

Advanced Smart Contract Development

[Profile](#)

2025

- Completed Blockchain Basics, Foundry Fundamentals, Advanced Foundry.